DS-IB Net: Ultra-Lightweight Weakly-Supervised Video Anomaly Detection through Synergistic Dual Streams and Information Bottleneck

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Abstract

Weakly-Supervised Video Anomaly Detection (WS-VAD) primarily struggles to robustly model normal temporal dynamics and sensitively detect sparse anomalies using only video-level labels. Current methods often inadequately represent normal patterns or are too computationally costly due to complex architectures, hindering practical use. We introduce DSIB-Net, a lightweight, dual-stream Information Bottleneck (IB) based framework. Its core Dual-Output Information Bottleneck Encoder (DOIBE), driven by a Temporal Information Bottleneck Loss (TIBL), operationalizes IB. DOIBE produces a predictive representation ($\mathbf{z}_{\text{pred}}$) for dynamic modeling and a compression control vector ($\mathbf{z}_{\text{comp}}$) for redundancy compression, with TIBL assigning them distinct functional roles. Since IB compression in DOIBE discards fine-grained appearance details, a parallel Instantaneous Mode Monitoring Path (IMP) provides crucial compensation by extracting these neglected frame-level features. An Adaptive Feature Arbitration (AFA) module then intelligently fuses these streams, dynamically weighting $\mathbf{z}_{\text{pred}}$ based on IMP's appearance cues to balance robustness and sensitivity. Experiments show DSIB-Net delivers highly competitive detection performance with minimal computational overhead, offering an efficient and practical WS-VAD solution.The code and dataset for the model are open-source and available at https://github.com/modadundun/DSIB-Net.

CCS Concepts

• **Computing methodologies → Scene anomaly detection**.

ACM Reference Format:
Tao Zhu, Qi Yu, Heran Song, Yuheng Cheng, Shiyu Li, Yue Liu, Xinyi Tu, and Kaiwen Luo. 2025. DS-IB Net: Ultra-Lightweight Weakly-Supervised Video Anomaly Detection through Synergistic Dual Streams and Information Bottleneck. In *ACM Multimedia Asia (MMAsia '25), December 09–12, 2025, Kuala Lumpur, Malaysia.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3743093.3771006

1 Introduction

The primary bottleneck in Weakly-Supervised Video Anomaly Detection (WS-VAD) arises from a challenging scenario[4, 13, 16, 17]. Video sequences often span tens of thousands of frames, yet models are provided only with a single video-level label[11, 27, 33]. This makes it difficult for models to precisely distinguish at the frame level between the inherent complex dynamics of normal activities and sparse, diverse anomalies[18, 19, 31, 32]. This bottleneck directly leads to two common deficiencies in existing models. First, there is a lack of robustness to atypical variations within normal activities. Such normal events are often misidentified as anomalies, resulting in high false alarm rates. Second, complex network architectures, designed to pursue detection performance for weak signals, tend to be computationally expensive. This hinders their efficient deployment in resource-constrained real-world scenarios. In recent years, new approaches have emerged, such as UMIL[12], transformers[7, 28], diffusion models[8, 21, 26], multi-modal solutions[10, 15, 25], and dual-branch architectures combining global and local information[15, 25]. While these methods have

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MMAsia '25, Kuala Lumpur, Malaysia
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ACM ISBN 979-8-4007-2005-5/25/12
https://doi.org/10.1145/3743093.3771006

effectively improved detection accuracy, their computational complexity has sharply increased. Consequently, they face challenges in meeting practical demands.

To systematically address these challenges, this paper introduces DSIB-Net. The goal is to achieve an effective balance between robustness and sensitivity under weak labels while significantly reducing model complexity. DSIB-Net is an ultra-lightweight dual-stream framework built upon the Information Bottleneck (IB)[5, 20, 23] principle. The core strategy of DSIB-Net involves several key actions. First, it distills essential, robust, and highly compressed predictive representations of normal temporal dynamics through lossy compression guided by the IB principle. Simultaneously, it establishes a parallel, dedicated path to capture instantaneous appearance structural information. This information is inevitably ignored during the aforementioned compression process but is crucial for identifying specific (especially instantaneous) anomalies. Finally, it adaptively fuses these two types of information through an intelligent arbitration mechanism.

The key to the framework is the Dual-Output Information Bottleneck Encoder (DOIBE). Under the strict constraint of a specially designed Temporal Information Bottleneck Loss (TIBL), DOIBE creatively materializes the IB principle into two functionally distinct and complementary latent variable components. The predictive representation ($\mathbf{z}_{\text{pred}}$) focuses on accurately encoding and predicting the core temporal evolution of normal activities. This aims to construct a stable and generalizable model of normal dynamics. Meanwhile, the compression control vector ($\mathbf{z}_{\text{comp}}$) collaborates with $\mathbf{z}_{\text{pred}}$ to achieve extreme compression of redundant input information irrelevant to predicting normal dynamics. This is achieved by strictly controlling the information complexity of the overall latent representation, thereby enhancing representation robustness and avoiding overfitting to superficial variations. The minimization process of TIBL encourages the model to actively distinguish between information needed for prediction and information to be discarded for compression during learning.

Although DOIBE achieves efficient predictive modeling of normal dynamics through IB compression, this process inevitably selectively ignores or smooths out certain features. These are instantaneous appearance features that may not have significant dynamic predictive value but could structurally indicate anomalies. This strategic abandonment of information is the direct motivation for designing the Instantaneous Mode Monitoring Path (IMP). IMP, as a lightweight parallel module, is specifically responsible for compensating for this lost local, instantaneous appearance information during DOIBE's compression process. IMP extracts raw, frame-level appearance patterns. This provides the model with a supplementary perspective on the current visual scene's immediate state, independent of dynamic analysis.

We further designed the Adaptive Feature Arbitration (AFA) module to effectively utilize and intelligently balance information from DOIBE and IMP. DOIBE provides compressed dynamic information, while IMP supplies raw instantaneous appearance information. AFA learns to interpret the evidence provided by IMP's output of instantaneous appearance features. Based on this interpretation, it dynamically adjusts the contribution weight of DOIBE's $\mathbf{z}_{\text{pred}}$ in the final fused feature. This mechanism enables DSIB-Net to intelligently focus on the more indicative information source in different video contexts, such as dynamic anomalies with normal appearance, or abnormal appearance with stable dynamics. Consequently, this significantly enhances sensitivity to various types of anomalies, especially instantaneous or appearance-based ones, while maintaining robustness to complex normal dynamics.

The main contributions of this paper are summarized as follows:

- IMP compensates for appearance features discarded by DOIBE's IB compression, enhancing sensitivity to instantaneous anomalies.
- AFA intelligently fuses DOIBE and IMP streams by dynamically weighing appearance evidence to balance robustness and sensitivity.
- DSIB-Net, an integrated lightweight dual-stream architecture, tackles core WS-VAD challenges by balancing detection accuracy with computational efficiency.

2 Proposed Method

The overall architecture of DSIB-Net is shown in Figure 1. Its core consists of a dual-stream feature encoding path, an adaptive feature arbitration module, and a classification module. Specifically, it includes the Dual-Output Information Bottleneck Encoder (DOIBE) stream and the Instantaneous Mode Monitoring Path (IMP) stream. These are followed by the Adaptive Feature Arbitration (AFA) module and a final MLP classifier. The entire model is trained by optimizing a main loss $\mathcal{L}_{\text{main}}$ and a specially designed auxiliary loss, the Temporal Information Bottleneck Loss $\mathcal{L}_{\text{TIBL}}$.

2.1 Stream 1: Dual-Output Information Bottleneck Encoder (DOIBE)

The core objective of the DOIBE stream is to learn robust and highly compressed representations of normal activity temporal dynamics. This is achieved by strictly adhering to the Information Bottleneck principle. The input visual feature sequence $\mathbf{X} = \{\mathbf{x}_1, ..., \mathbf{x}_T\}$ is first preliminarily encoded for temporal information by a Gated Recurrent Unit f_{GRU}[2]. This process yields a hidden state sequence $\{\mathbf{h}_1, ..., \mathbf{h}_T\}$. Each hidden state is computed as $\mathbf{h}_t = f_{\text{GRU}}(\mathbf{x}_t, \mathbf{h}_{t-1}; \mathbf{W}_{\text{gru}})$.

Based on $\mathbf{h}_t$, DOIBE outputs its two core conceptual latent variable components: $\mathbf{z}_{\text{pred}}(t)$ and $\mathbf{z}_{\text{comp}}(t)$. The component $\mathbf{z}_{\text{pred}}(t)$ is responsible for encapsulating the core dynamic information needed to predict the next normal frame $\mathbf{x}_{t+1}$; its aim is to maximize $I(\mathbf{Z}_t; \mathbf{x}_{t+1})$. The component $\mathbf{z}_{\text{comp}}(t)$, in coordination with $\mathbf{z}_{\text{pred}}(t)$, aims to minimize $I(\mathbf{Z}_t; \mathbf{X}_{\leq t})$. It achieves this by regulating the overall information complexity of the latent variable $\mathbf{Z}_t$, leading to extreme compression of the input $\mathbf{X}_{\leq t}$.

To effectively optimize and obtain these two functional components in practice, we adopt a variational inference implementation.We parameterize the posterior distribution of the latent variable $\mathbf{Z}_t$, $q_t(\mathbf{Z}_t|\mathbf{X}_{\leq t})$, as a diagonal Gaussian distribution given by:

$$\mathcal{N}(\boldsymbol{\mu}_t, \text{diag}(\exp(\mathbf{logvar}_t))) \tag{1}$$

Within this variational framework:

- The function of $\mathbf{z}_{\text{pred}}(t)$ is primarily undertaken and realized by the mean vector $\boldsymbol{\mu}_t$ of this posterior distribution.
- The function of $\mathbf{z}_{\text{comp}}(t)$ is realized by controlling information complexity. This control is achieved through the

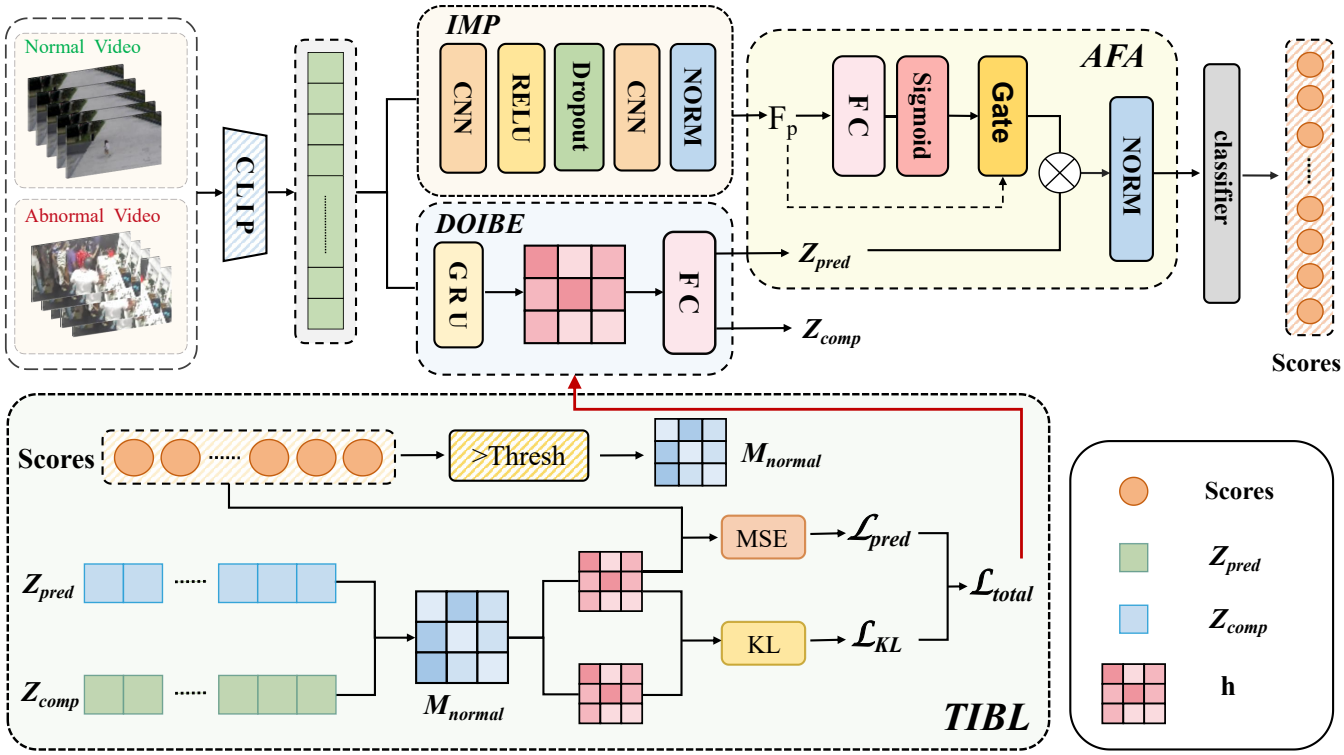

Figure 1: The overall framework of DSIB-Net, illustrating the dual-stream architecture with DOIBE and IMP paths, fused by the AFA module, and the role of TIBL.

regulation of the log-variance vector $\mathbf{logvar}_t$ (and thus the variance magnitude) of this posterior distribution.

These two parameterized vectors are obtained from $\mathbf{h}_t$ through independent linear mappings:

$$\boldsymbol{\mu}_t = f_\mu(\mathbf{h}_t; \mathbf{W}_\mu) \in \mathbb{R}^L \tag{2}$$

$$\mathbf{logvar}_t = f_{\text{logvar}}(\mathbf{h}_t; \mathbf{W}_{\text{logvar}}) \in \mathbb{R}^L \tag{3}$$

Here, L is the latent variable dimension, set to be the same as the input feature dimension D. The mean sequence $\mathbf{M} = \{\boldsymbol{\mu}_t\}$ serves as the dynamic prediction core extracted by DOIBE and is passed to the subsequent Adaptive Feature Arbitration (AFA) module. Concurrently, $\mathbf{M}$ and the log-variance sequence $\mathbf{LogVarSeq} = \{\mathbf{logvar}_t\}$ are jointly used for calculating and optimizing the Temporal Information Bottleneck Loss (TIBL).

2.2 Stream 2: Instantaneous Mode Monitoring Path (IMP)

The Instantaneous Mode Monitoring Path (IMP) aims to compensate for fine-grained, instantaneous appearance patterns that DOIBE inevitably discards during its Information Bottleneck (IB) guided compression. While DOIBE prioritizes robust temporal dynamic modeling and high compression, it may overlook or smooth out local, instantaneous appearance details if they contribute little to predicting normal dynamics or increase latent variable complexity. IMP is specifically designed to independently capture these strategically ignored raw, instantaneous frame-level appearance features.

IMP employs a lightweight one-dimensional convolutional network (f_{IMP}) that directly operates on raw input features $\mathbf{x}_t$ of each frame. The input feature sequence $\mathbf{X} \in \mathbb{R}^{B\times T\times D}$ is first permuted to $\mathbf{X}^\intercal \in \mathbb{R}^{B\times D\times T}$, allowing 1D convolution kernels to operate along the feature dimension D. Subsequently, a series of 1D convolutional layers process these features:

$$\mathbf{L}_{\text{permuted}} = f_{\text{IMP}}(\mathbf{X}^\intercal; \mathbf{W}_{\text{imp}}) \in \mathbb{R}^{B\times D\times T} \tag{4}$$

The network f_{IMP}, comprising convolution, ReLU activation, and Dropout layers, efficiently captures intra-frame statistics, local patterns, and instantaneous inter-channel correlations. The convolutional output $\mathbf{L}_{\text{permuted}}$ is rearranged to the time-dominant dimension and then processed through Layer Normalization (LN) to generate a stable, detail-rich sequence of instantaneous mode features $\mathbf{F}_{\text{p}} = \{\mathbf{f}_{\text{p},1}, \ldots, \mathbf{f}_{\text{p},T}\}$.

$$\mathbf{F}_{\text{p}} = \text{LN}(\text{Permute}(\mathbf{L}_{\text{permuted}})) \in \mathbb{R}^{B\times T\times D} \tag{5}$$

These features $\mathbf{f}_{\text{p},t}$ provide rich information about the instantaneous appearance patterns of the current frame at each time step t for the subsequent AFA module.

2.3 Adaptive Feature Arbitration Module (AFA)

The core mission of the AFA module is to act as an intelligent arbiter and information integrator. It operates between two information

sources: the dynamic predictive representation from the DOIBE stream (realized as $\boldsymbol{\mu}_t$, representing $\mathbf{z}_{\text{pred}}(t)$), refined by the IB principle, and the raw instantaneous appearance mode features $\mathbf{f}_{\text{p},t}$ from the IMP stream. AFA dynamically adjusts the contribution of DOIBE's representation based on the grounds of the current frame's instantaneous appearance patterns provided by IMP. This achieves an adaptive balance between model robustness and sensitivity.

Specifically, the instantaneous mode features $\mathbf{f}_{\text{p},t}$ from IMP are first fed into a gating function f_g. This function consists of a linear layer followed by a Sigmoid activation function σ. It maps $\mathbf{f}_{\text{p},t}$ to a gating vector $\mathbf{g}_t \in [0,1]^D$, which has the same dimensionality as the features:

$$\mathbf{g}_t = \sigma(f_g(\mathbf{f}_{\text{p},t}; \mathbf{W}_g)) \tag{6}$$

Each element value of this gating vector $\mathbf{g}_t$ represents the confidence level or modulation strength for the corresponding dimension of DOIBE's representation $\boldsymbol{\mu}_t$. This representation is based on AFA's interpretation of IMP's appearance evidence.

Subsequently, the computed gating vector $\mathbf{g}_t$ is used to element-wise modulate the dynamic prediction core $\boldsymbol{\mu}_t$ from DOIBE:

$$\hat{\boldsymbol{\mu}}_t = \boldsymbol{\mu}_t \odot \mathbf{g}_t \tag{7}$$

This multiplicative modulation implements an intelligent arbitration. During model training, AFA learns to interpret $\mathbf{f}_{\text{p},t}$. If $\mathbf{f}_{\text{p},t}$ indicates normal appearance patterns, the element values of $\mathbf{g}_t$ output by f_g approach 1, and DOIBE's $\boldsymbol{\mu}_t$ dominates the fusion result. Conversely, if $\mathbf{f}_{\text{p},t}$ indicates potential statistical rarity or significant deviations related to anomalies in the appearance patterns, the element values of $\mathbf{g}_t$ output by f_g approach 0. This weakens the inertial predictions of $\boldsymbol{\mu}_t$ based on historical normal dynamics, allowing the instantaneous appearance deviation signals captured by IMP to dominate the fused features.

Finally, the adaptively modulated features $\hat{\boldsymbol{\mu}}_t$ are processed through Layer Normalization (LN). This yields the final fused features $\mathbf{f}_t \in \mathbb{R}^D$ used for classification, forming the feature sequence $\mathbf{F} = \{\mathbf{f}_1, ..., \mathbf{f}_T\}$:

$$\mathbf{f}_t = \text{LN}(\hat{\boldsymbol{\mu}}_t) \tag{8}$$

This adaptive fusion strategy ensures that the final representation can dynamically synergize DOIBE's dynamic information and IMP's instantaneous appearance information.

2.4 Temporal Information Bottleneck Loss (TIBL)

TIBL is an auxiliary loss function. It is designed to optimize the two core conceptual components output by the DOIBE module: $\mathbf{z}_{\text{pred}}$ and $\mathbf{z}_{\text{comp}}$. Its goal is to ensure that the latent variables learned by DOIBE strictly adhere to the Information Bottleneck principle. The TIBL calculation is performed only on the set of video frames $\mathcal{T}_{\text{normal}}$ that the model dynamically determines to be normal. These normal frames are selected based on the frame-wise anomaly scores s_t output by the main classifier (defined in Section 2.5). A frame is included in $\mathcal{T}_{\text{normal}}$ only when its score s_t is below a preset normal threshold, normal_thresh. This selection process is implemented by generating a binary mask $\mathbf{M}_{\text{normal}}$, where an element $m_t = 1$ if the anomaly score $s_t <$ normal_thresh, and $m_t = 0$ otherwise. The set $\mathcal{T}_{\text{normal}}$ thus consists of frames for which $m_t = 1$.This selective strategy ensures that TIBL focuses on learning and compressing normal patterns from high-confidence normal samples, thereby avoiding interference from anomalous data. TIBL consists of the following two core components:

1. **Prediction Loss ($\mathcal{L}_{\text{pred}}$)**: This loss term aims to maximize the latent variable's ability to predict future normal dynamics. It specifically optimizes DOIBE's output $\mathbf{z}_{\text{pred}}(t)$, whose function is primarily undertaken by $\boldsymbol{\mu}_t$. The loss employs a small predictor network $p(\cdot; \mathbf{W}_p)$. This network predicts the current raw normal visual feature $\mathbf{x}_t$ using the mean representation $\boldsymbol{\mu}_{t-1}$ from the previous time step. The ability of DOIBE to generate predictive $\boldsymbol{\mu}_t$ and the parameters of the predictor p are enhanced by minimizing the Mean Squared Error (MSE) between the predicted value $p(\boldsymbol{\mu}_{t-1})$ and the true target $\mathbf{x}_t$:

$$\mathcal{L}_{\text{pred}} = \mathbb{E}_{(b,t-1)\in\mathcal{T}_{\text{pred}}}[\text{MSE}(\mathbf{x}_{b,t}, p(\boldsymbol{\mu}_{b,t-1}; \mathbf{W}_p))] \tag{9}$$

where $\mathcal{T}_{\text{pred}}$ is the subset of frames in $\mathcal{T}_{\text{normal}}$ suitable for prediction; it excludes the first frame of the sequence.

2. **KL Divergence Loss ($\mathcal{L}_{\text{KL}}$)**: This loss term aims to minimize the information complexity encoded by the latent variable. This achieves the compression goal of the IB principle and optimizes DOIBE's output $\mathbf{z}_{\text{comp}}(t)$. The function of $\mathbf{z}_{\text{comp}}(t)$ is realized through the regulation of $\mathbf{logvar}_t$ and its coordination with $\boldsymbol{\mu}_t$. Within the variational framework, this minimization is achieved by minimizing the KL divergence. This divergence is between the latent variable's posterior distribution $q_t(\mathbf{Z}_t|\mathbf{X}_{\leq t})$ and a standard Gaussian prior $p(\mathbf{Z}) = \mathcal{N}(\mathbf{Z}|0, \mathbf{I})$. The posterior distribution, parameterized by $\boldsymbol{\mu}_t$ and $\mathbf{logvar}_t$, is defined as:

$$q_t(\mathbf{Z}_t|\mathbf{X}_{\leq t}) = \mathcal{N}(\mathbf{Z}_t|\boldsymbol{\mu}_t, \text{diag}(\exp(\mathbf{logvar}_t))) \tag{10}$$

$$\begin{aligned}\mathcal{L}_{\text{KL}} &= \mathbb{E}_{(b,t)\in\mathcal{T}_{\text{normal}}}[\text{KL}(q_t(\mathbf{Z}_{b,t}|\mathbf{X}_{b,\leq t})||p(\mathbf{Z}_{b,t}))] \\ &= \mathbb{E}_{(b,t)\in\mathcal{T}_{\text{normal}}}\Bigg[-\frac{1}{2}\sum_{j=1}^{L}(1 + \mathbf{logvar}_{b,t,j} \\ &\qquad -\boldsymbol{\mu}_{b,t,j}^2 - \exp(\mathbf{logvar}_{b,t,j}))\Bigg]\end{aligned} \tag{11}$$

Minimizing this KL divergence forces the parameterized posterior distribution q_t to approach the simple prior. This effectively penalizes complex representations that attempt to encode too many specific details of $\mathbf{X}_{\leq t}$, which would cause q_t to deviate from the prior. This process compels the encoder to discard redundant information, making $\mathbf{Z}_t$ a more concise and generalized representation.

The final TIBL loss is a weighted sum of these two core components:

$$\mathcal{L}_{\text{TIBL}} = \mathcal{L}_{\text{pred}} + \beta \cdot \mathcal{L}_{\text{KL}} \tag{12}$$

By jointly minimizing $\mathcal{L}_{\text{pred}}$ and $\mathcal{L}_{\text{KL}}$, TIBL guides DOIBE to learn a specific type of latent variable representation. This representation can both effectively predict normal dynamics (corresponding to optimizing $\mathbf{z}_{\text{pred}}$) and highly compress irrelevant information (corresponding to optimizing $\mathbf{z}_{\text{comp}}$). The hyperparameter β balances these two core IB objectives: prediction accuracy and compression strength.

2.5 Loss Functions

The main loss $\mathcal{L}_{\text{main}}$ uses a Multiple Instance Learning (MIL)[14, 22] strategy. From each video i, K_{mil} key frames most indicative of its label Y_i are selected. This selection is based on their frame-wise

anomaly scores s_t. $\mathcal{L}_{\text{main}}$ is the mean Binary Cross-Entropy (BCE) loss between the anomaly logits $\ell^*_{i,j}$ of these selected frames and their video-level labels:

$$\mathcal{L}_{\text{main}} = \frac{1}{N_{selected}} \sum_{i,j} \text{BCE}(Y_i, \ell^*_{i,j}) \tag{13}$$

Here, $N_{selected}$ is the total count of such selected frames in a batch. Frame-wise anomaly scores are calculated as $s_t = \sigma(\ell_t)$, where ℓ_t are raw logits from the MLP classifier and $\sigma(\cdot)$ is the Sigmoid function. These scores s_t are used for MIL selection, TIBL's normal frame selection, and final anomaly determination. The BCE loss is defined as $\text{BCE}(Y, \ell^*) = -[Y \log \sigma(\ell^*) + (1 - Y) \log(1 - \sigma(\ell^*))]$.

The training of the model in this paper aims to minimize a combined loss function $\mathcal{L}_{\text{total}}$. This function is a weighted sum of the main loss term $\mathcal{L}_{\text{main}}$ and the auxiliary Temporal Information Bottleneck Loss $\mathcal{L}_{\text{TIBL}}$ (detailed in Section 2.4):

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + \lambda_{\text{TIBL}} \cdot \mathcal{L}_{\text{TIBL}} \tag{14}$$

where λ_{TIBL} is a hyperparameter that balances the contributions of the two loss components.

3 Experiments

3.1 Datasets & Implementation Details

Datasets and Evaluation Metrics: Experiments are conducted on the ShanghaiTech and UCF-Crime benchmark datasets, using the standard frame-level Area Under the Curve (AUC) as the core performance metric.

Implementation Details: Visual features are extracted using a pre-trained ViT-B/16 model [25]. Model training employs the Adam optimizer (learning rate 10^{-4}) with batches of 30 positive and 30 negative samples. The MIL-based main loss ($\mathcal{L}_{\text{main}}$) uses $K_{mil} = 5$. The auxiliary Temporal Information Bottleneck Loss ($\mathcal{L}_{\text{TIBL}}$, weight $\lambda_{\text{TIBL}} = 0.1$, Eq. 14) includes a KL divergence term (weight $\beta = 0.01$, Eq. 12) and uses normal_thresh = 0.5 for selecting normal frames for its calculation.

3.2 Comparison with SOTA Methods

Detection Performance: Table 1 clearly presents a performance comparison of DSIB-Net with representative SOTA methods in the field on the ShanghaiTech and UCF-Crime datasets. DSIB-Net achieves an AUC of 98.13% on the ShanghaiTech dataset and 87.65% on the UCF-Crime dataset. These results conclusively demonstrate the outstanding detection performance of DSIB-Net, matching or surpassing existing top-tier methods on mainstream benchmarks.

Computational Efficiency: Table 2 compares DSIB-Net with some representative SOTA methods in terms of computational resource consumption. DSIB-Net demonstrates significant advantages in key efficiency metrics such as model parameter count (0.005G), disk size (17.7MB), which are considerably lower than most existing high-performance models.

3.3 Ablation Study

We performed ablation studies (Table 3) to evaluate each DSIB-Net component's contribution and synergy.

Table 3 shows the impact of each component. First, DOIBE with TIBL significantly improved results over the baseline and IMP-only models. This validates its IB-guided learning for dynamic representation and compression. Second, IMP consistently boosted performance. It uses instantaneous appearance features to compensate for DOIBE's compression losses, enhancing sensitivity to certain anomalies. Third, the full DSIB-Net with AFA achieved the best performance. Its intelligent fusion surpassed simple additive fusion. This confirms the framework's robust design and crucial module synergy.

Table 1: Comparison of detection performance (%) and feature extractors of DSIB-Net with SOTA methods on ShanghaiTech (SH-AUC) and UCF-Crime (UCF-AUC). The best results are marked in bold, and the second best are underlined.

Method	Feature	SH-AUC (%)	UCF-AUC (%)
Sultani et al. [18]	C3D	85.33	77.92
Zhong et al. [31]	C3D	76.44	81.08
CLAWS [29]	C3D	89.67	83.03
MIST [3]	I3D	94.83	82.03
RTFM [19]	I3D	97.21	84.30
MSL [6]	I3D	95.45	85.30
S3R [24]	I3D	97.48	85.99
DAR [9]	I3D	97.54	85.18
Cho et al. [1]	I3D	97.60	86.01
CUPL [30]	I3D	–	86.22
VadCLIP [25]	CLIP	97.49	**88.02**
DSIB-Net (Ours)	CLIP	**98.13**	87.65

Table 2: Computational efficiency (lower is better).

Method	Parameters (G)	Size (MB)	Test Time (s)
MIST [3]	0.030	48.5	0.0059
RTFM [19]	0.020	94.3	0.0636
S3R [24]	0.050	310.7	0.0370
VadCLIP [25]	0.035	619.1	0.0410
DSIB-Net	**0.005**	**17.7**	**0.0009**

Table 3: Ablation study results for DSIB-Net.

Model Configuration	SH-AUC (%)	UCF-AUC (%)
(1) Baseline (GRU + Classifier)	95.53	83.77
(2) +IMP	97.18	84.05
(3) +DOIBE/TIBL	97.72	85.11
(4) +DOIBE+IMP	97.95	86.83
(5) DSIB-Net (Full Model)	**98.13**	**87.65**

3.4 Key Hyperparameter Sensitivity Analysis

We investigated the impact of two key hyperparameters in the TIBL loss—the total TIBL weight λ_{TIBL} and the KL divergence term weight β—on model performance. As shown in Table 4, when these hyperparameters are adjusted within a certain range around their default values ($\lambda_{\text{TIBL}} = 0.1, \beta = 0.01$, which yield our best reported

results), DSIB-Net's performance exhibits good robustness with minor AUC fluctuations relative to this new optimum. This indicates that the model's performance is not extremely sensitive to the precise selection of these key hyperparameters, reducing the complexity of parameter tuning in practical applications.

Table 4: Sensitivity analysis of DSIB-Net key hyperparameters λ_{TIBL} and β (SH-AUC %, UCF-AUC %). Default values leading to best results are marked with *.

λ_{TIBL} variation ($\beta = 0.01$)			β variation ($\lambda_{\text{TIBL}} = 0.1$)		
Value	SH-AUC (%)	UCF-AUC (%)	Value	SH-AUC (%)	UCF-AUC (%)
0.01	97.80	87.31	0.001	97.96	87.45
0.05	97.99	87.52	0.005	98.03	87.58
0.1*	**98.13**	**87.65**	0.01*	**98.13**	**87.65**
0.5	97.91	87.39	0.05	97.77	87.23
1.0	97.67	87.15	0.1	97.49	86.99

3.5 Feature and Qualitative Results Analysis

Fused Feature Visualization: UMAP visualization of the AFA module's final fused features, $\mathbf{f}_t$, demonstrates their strong discriminative power (Figure 2). On both ShanghaiTech and UCF-Crime test sets, features from normal (blue triangles) and anomalous (red asterisks) frames form clearly distinct and separable clusters. This visually confirms DSIB-Net's ability to learn discriminative representations, supporting its excellent detection performance.

Qualitative Detection Results: Frame-wise anomaly score curves from DSIB-Net on test samples (Figure 3) effectively highlight its detection capabilities. Scores consistently remain low during normal activities, but rise sharply to pinpoint anomalous events, then decrease post-event. These results visually confirm DSIB-Net's proficiency in distinguishing normal from anomalous activities and accurately timing them.

4 Conclusion

Our lightweight dual-stream framework for weakly-supervised video anomaly detection, DSIB-Net, delivers top-tier detection accuracy with remarkable operational efficiency. Compared to a recent SOTA method like VadCLIP, DSIB-Net reduces parameters by 85.7%, model size by 97.1%, and inference time by 97.8%. This combination of high accuracy and extreme efficiency highlights its strong potential for real-world applications. Future work will explore extending its core principles to other temporal analysis tasks.

Acknowledgments

This work was supported by the 2025 Jiangxi Graduate Innovation Program (Grant No. YC2025-S470).

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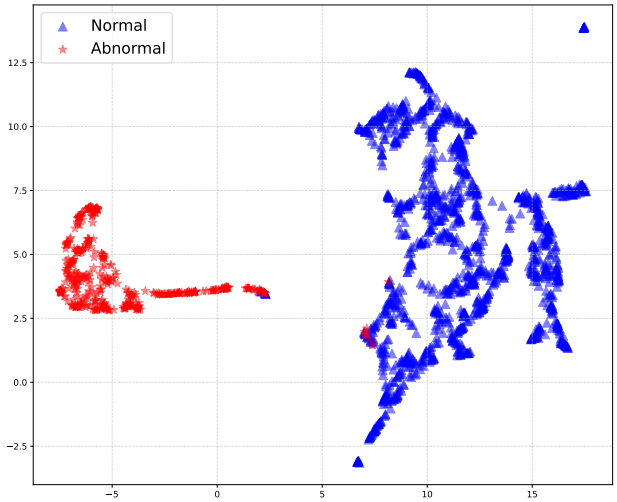
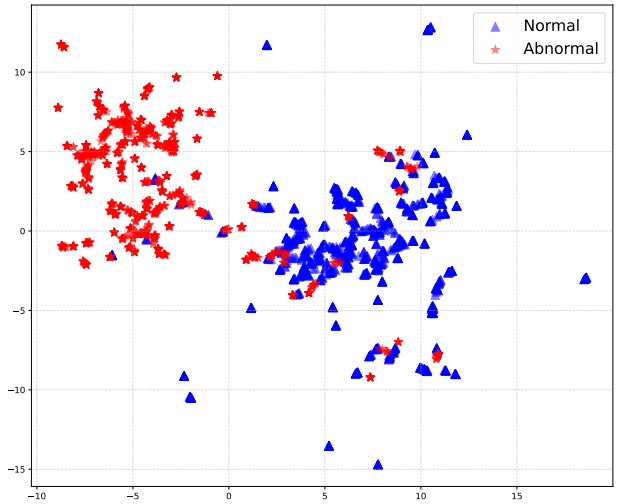


(a) ShanghaiTech dataset



(b) UCF-Crime dataset

Figure 2: UMAP visualization of DSIB-Net's final fused features.

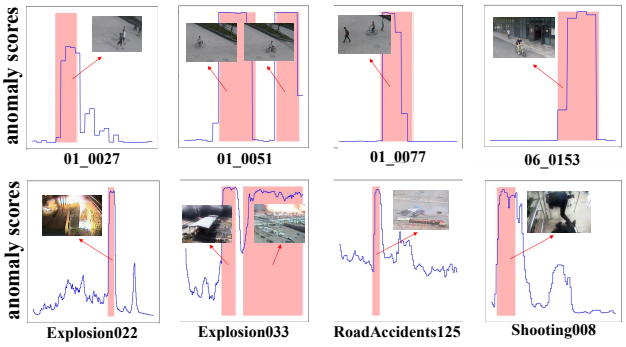


Figure 3: Sample frame-wise anomaly scores from DSIB-Net.

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